Math 257 Definitions and Theorems

Week 3: Modules 9-11

Fall 2023

9 Elementary Matrices

Definition 9.1. An **elementary matrix** is one that is obtained by performing a single elementary row operation on an identity matrix.
A **permutation matrix** is one that is obtained by performing row exchanges on an identity matrix.

Theorem 9.1. *If an elementary row operation is performed on an $m \times n$-matrix A, the resulting matrix can be written as EA, where the $m \times m$-matrix E is created by performing the same row operations on I_m.*

Theorem 9.2. *Let A, B be two $m \times n$-matrices and row-equivalent. Then there is a sequence $m \times m$-elementary matrices $E_1, \ldots, E_\ell$ such that*

$$E_\ell \ldots E_1 A = B.$$

10 Inverse of a Matrix

Remark. The *inverse* of a real number a is denoted by a^{-1}. For example, $7^{-1} = 1/7$ and $7 \cdot 7^{-1} = 7^{-1} \cdot 7 = 1$. Note that not all real numbers have inverse. Namely, 0^{-1} is not defined as there is no real number b such that $0 \cdot b = 1$.

Definition 10.1. An $n \times n$ matrix A is said to be **invertible** if there is an $n \times n$ matrix C satisfying

$$CA = AC = I_n$$

where I_n is the $n \times n$ identity matrix. We call C the **inverse** of A.

Theorem 10.1. *Let A be an invertible matrix, then its inverse C is unique.*

Theorem 10.2. *Suppose A and B are invertible. Then:*

(a) A^{-1} *is invertible and* $\left(A^{-1}\right)^{-1} = A$ *(i.e. A is the inverse of A^{-1}).*

(b) AB *is invertible and* $(AB)^{-1} = B^{-1}A^{-1}$.

(c) A^T *is invertible and* $\left(A^T\right)^{-1} = \left(A^{-1}\right)^T$.

Theorem 10.3. *Let A be an invertible $n \times n$ matrix. Then for each $\mathbf{b}$ in $\mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.*

11 Computing the Inverse

Theorem 11.1. *Let* $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. *If* $ad - bc \neq 0$, *then* A *is invertible and*

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

If $ad - bc = 0$, *then* A *is not invertible.*

Theorem 11.2. *Let* A *be an* $n \times n$*-matrix. The following are equivalent:*

- A *is invertible.*
- *the reduced echelon form of* A *is* I_n.

Theorem 11.3. *Suppose* A *is invertible. The every sequence of elementary row operations that reduces* A *to* I_n *will also transform* I_n *to* A^{-1}.

Algorithm 11.1.

- Place A and I side-by-side to form an augmented matrix $[\ A \mid I\ ]$. This is an $n \times 2n$ matrix (Big Augmented Matrix), instead of $n \times (n + 1)$.
- Perform row operations on this matrix (which will produce identical operations on A and I).
- By Theorem:

  $$[\ A \mid I\ ] \text{ will row reduce to } \left[\ I \mid A^{-1}\ \right]$$

  or A is not invertible.